



How to Achieve a Remarkable 35% Growth in Profit

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Executive Summary

Purpose of the Report

This report provides a comprehensive overview of sales and profit trends across various countries with a specific emphasis on the United States. It identifies key challenges and unprofitable elements such as high discount percentages and excessive shipping costs. Additionally, the report proposes several effective strategies to improve financial performance. The objective of this report is to analyze the business problems and implement solutions to enhance overall profit by up to 35%

Key Findings

1. The United States generates the highest sales and profits compared to other countries.
2. Texas, a state in the United States, experiences losses due to underperforming products which demonstrates high sales do not ensure high profit.
3. Binders and appliances are the major sources of losses that seriously affect the overall profitability in Texas.
4. High discounts and excessive shipping costs are identified as primary factors that negatively impact profit margins, particularly in Texas.
5. After reducing discounts and excessive shipping costs, overall profit in the United States can potentially increase by 35%.

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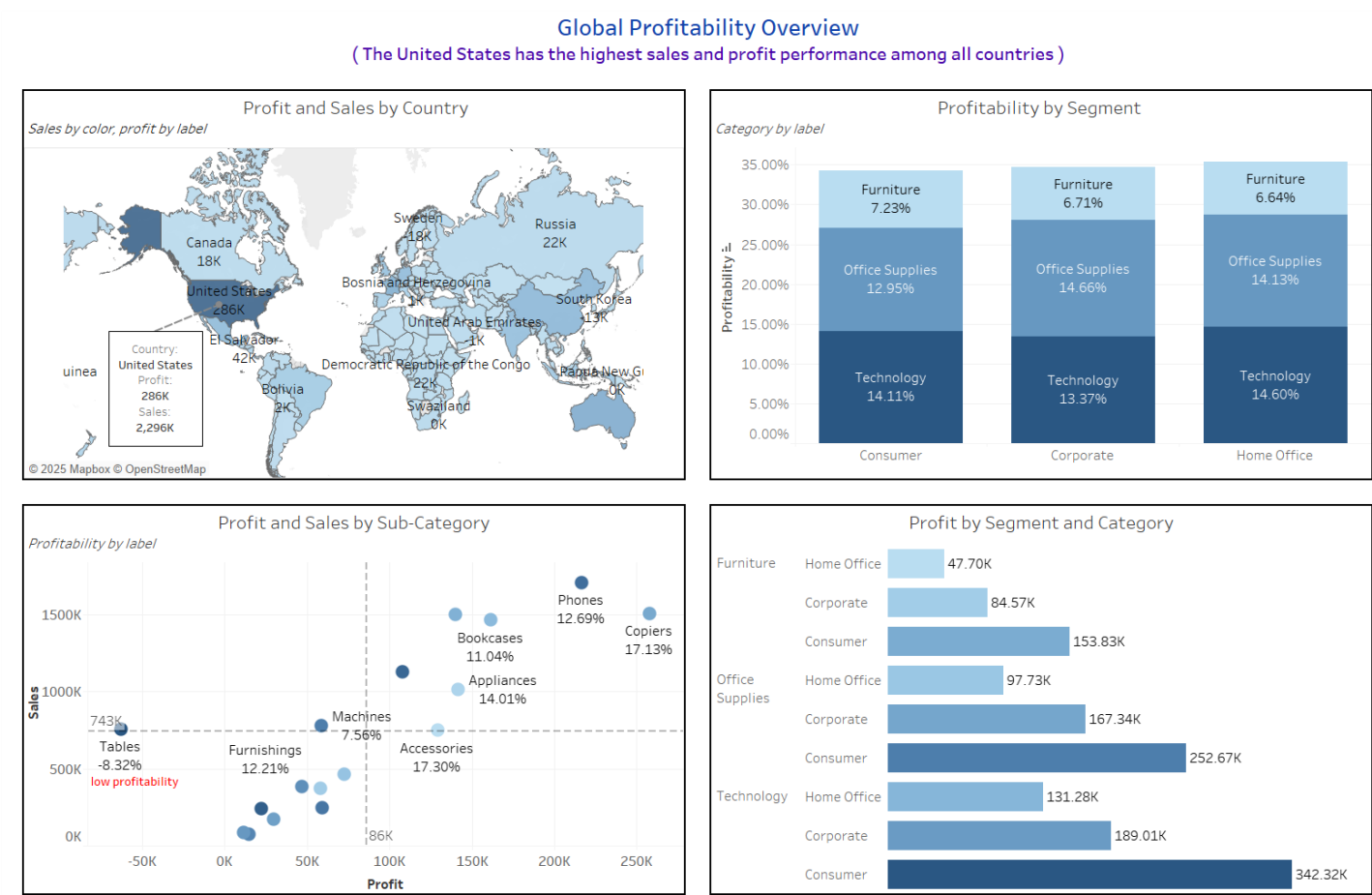
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Introduction

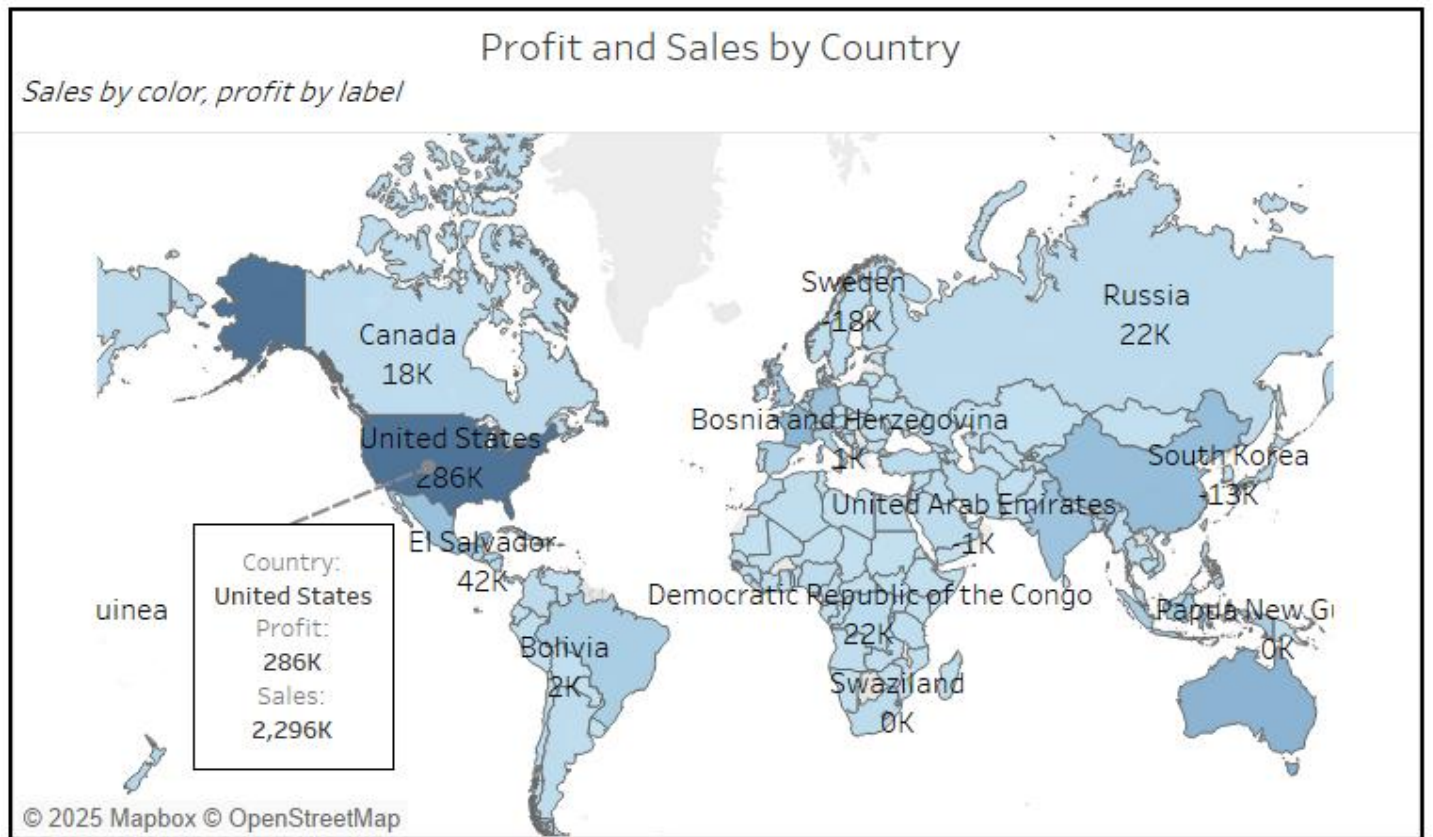
This report provides a comprehensive overview of sales and profit trends across regions, product categories and customer segments with a focus on the United States. It emphasizes that high sales do not always result in high profits. For instance, Texas suffers tremendous losses despite high sales due to non-profitable products like appliances and binders. The report also identifies the negative effects of excessive discount percentages and shipping costs which decrease profitability. However, key findings suggest that the reduction of discount percentages and optimization of shipping costs could lead to a 35% increase in profitability. In short, this Tableau visualization report presents feasible tactics to improve profit performance and operational efficiency.

Figure 1:



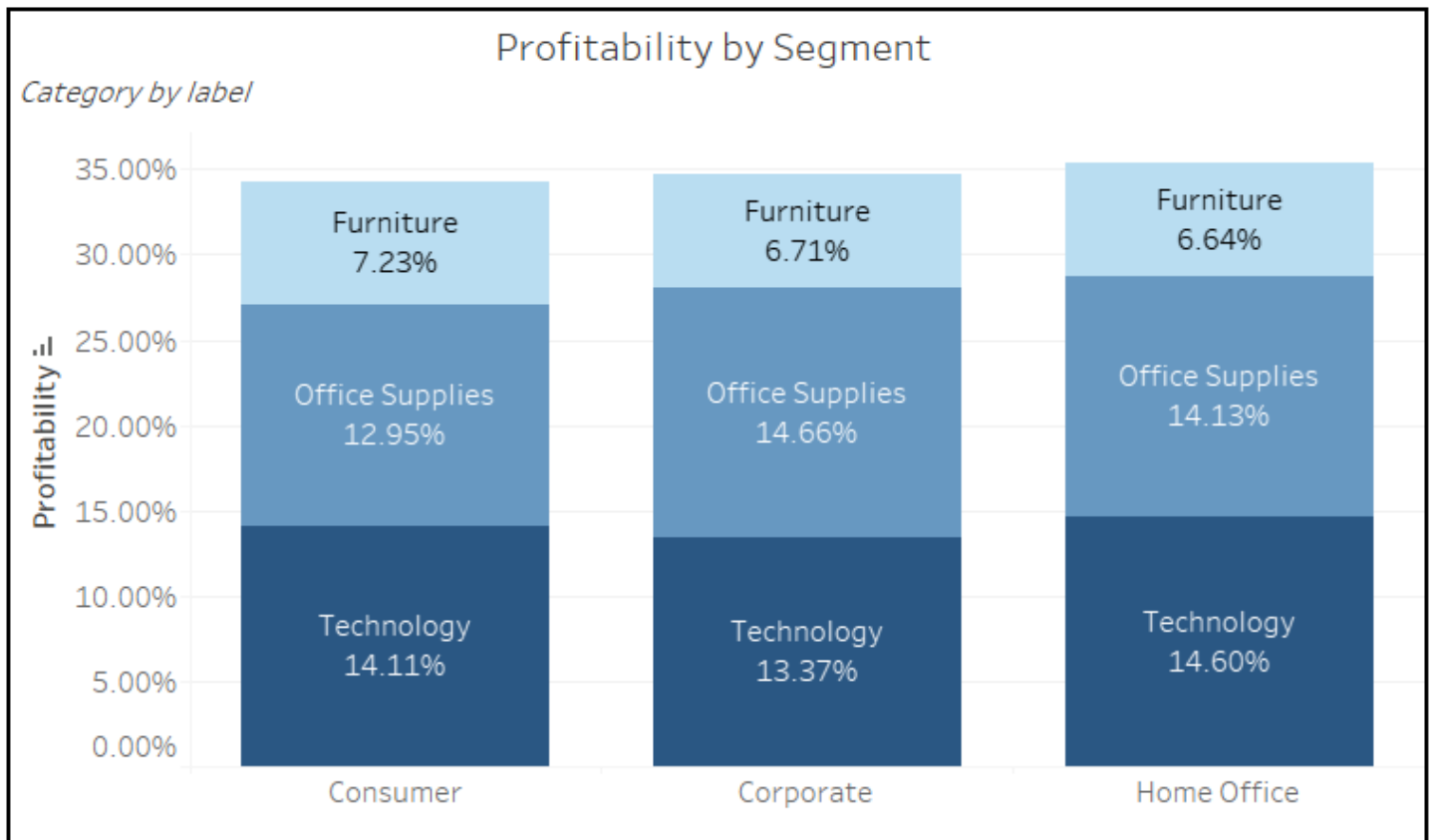
In the overview dashboard, it shows that the U.S. greatly outperformed other countries in global sales and profit which can be observed through the map graph organized by countries.

1.1 Global Profit and Sales Analysis



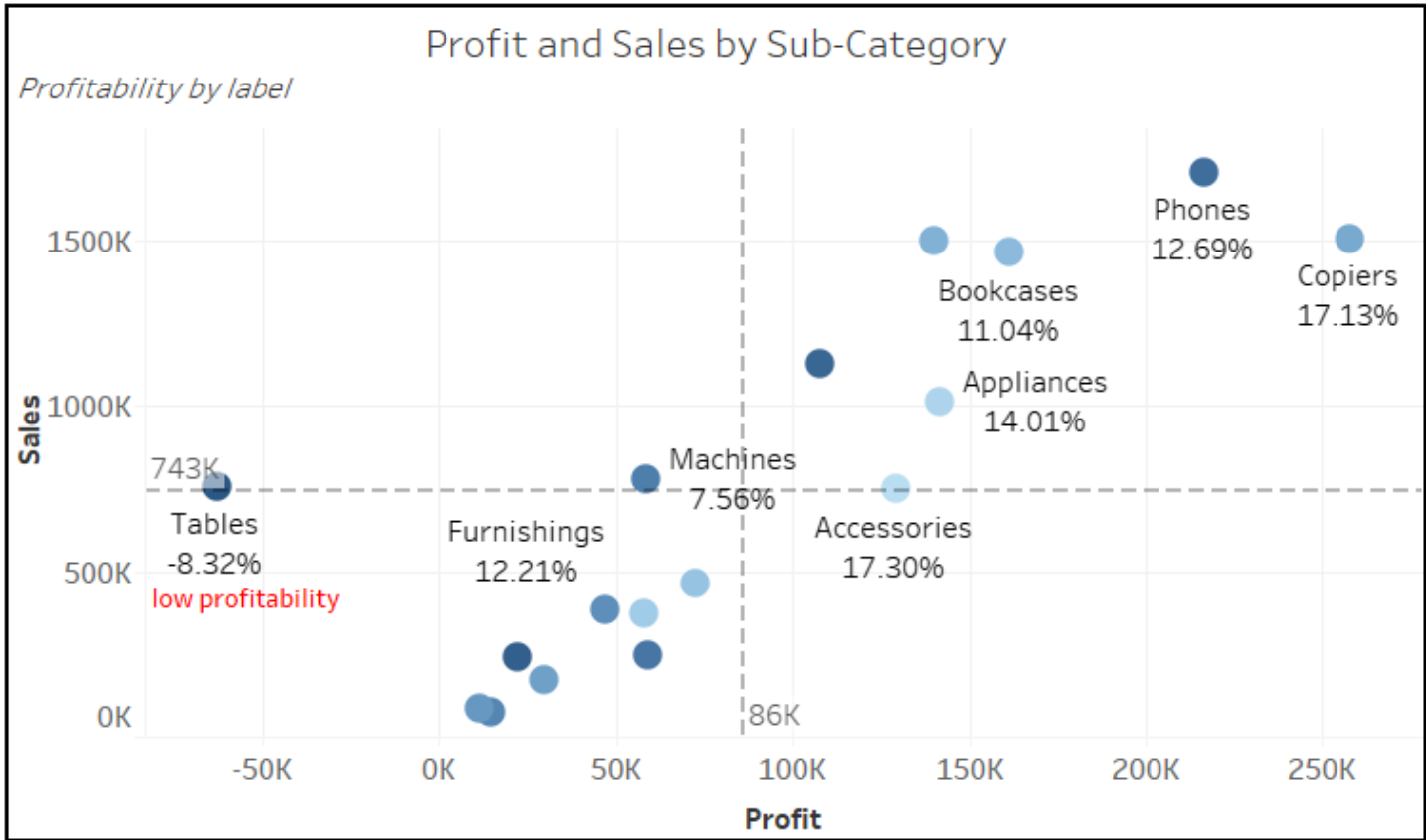
This map illustrates the worldwide distribution of sales and profit. The U.S. leads in both whereas some countries have negative profits. For example, Sweden (\$ -18K) and South Korea (\$-13K) suffer substantial losses. However, there is also a huge disparity between the revenue (\$286K) and sales volume (\$2296K) in the United States which shows potential inefficiency in profitability.

1.2 Profitability Analysis by Customer Segment



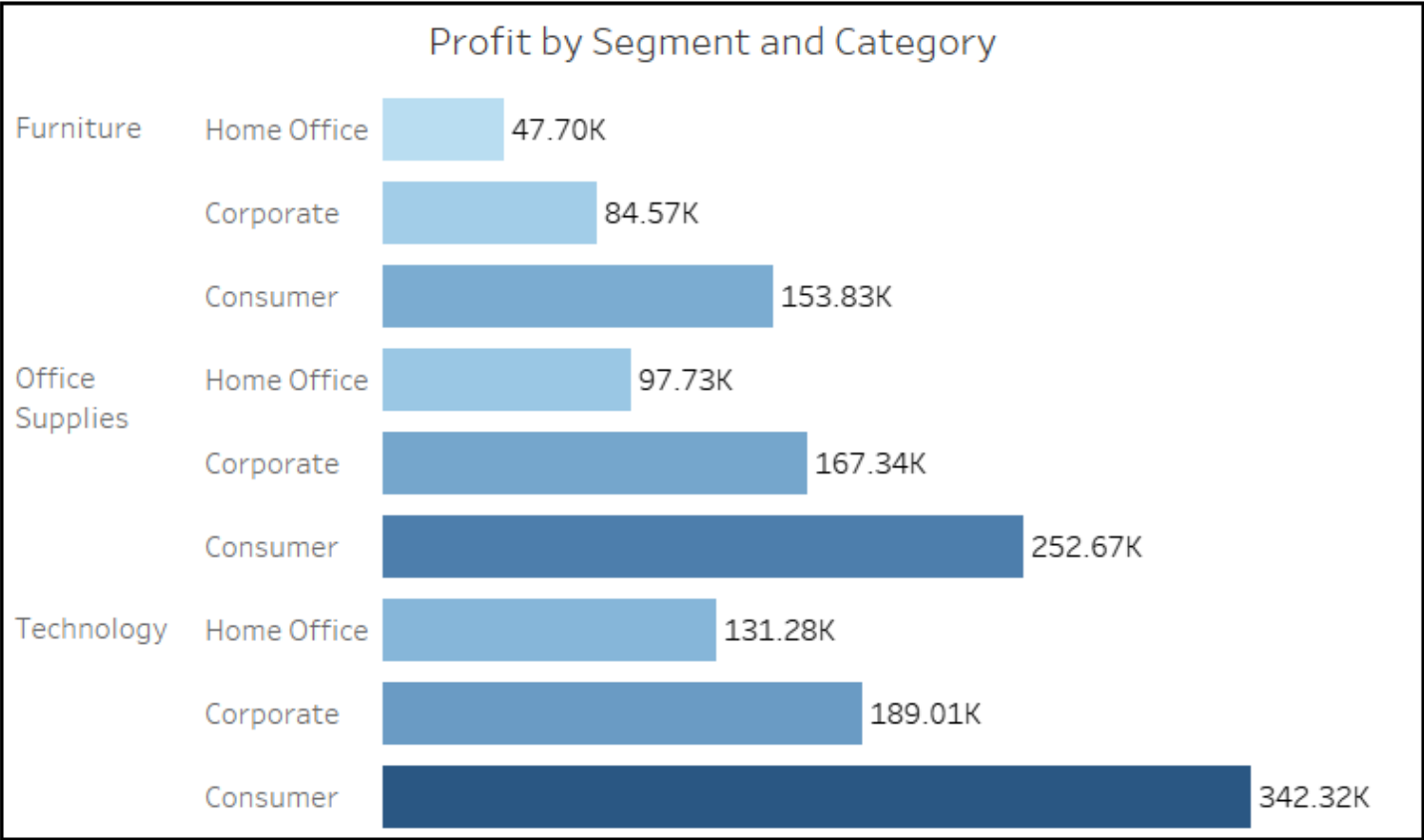
This stacked bar chart compares profitability percentages across three customer segments for product categories. Technology demonstrates the highest profitability in each segment while furniture consistently shows the lowest. This visualization highlights the relative performance of each product category and raises an issue about the furniture's low performance.

1.3 Product Profit and Sales Performance



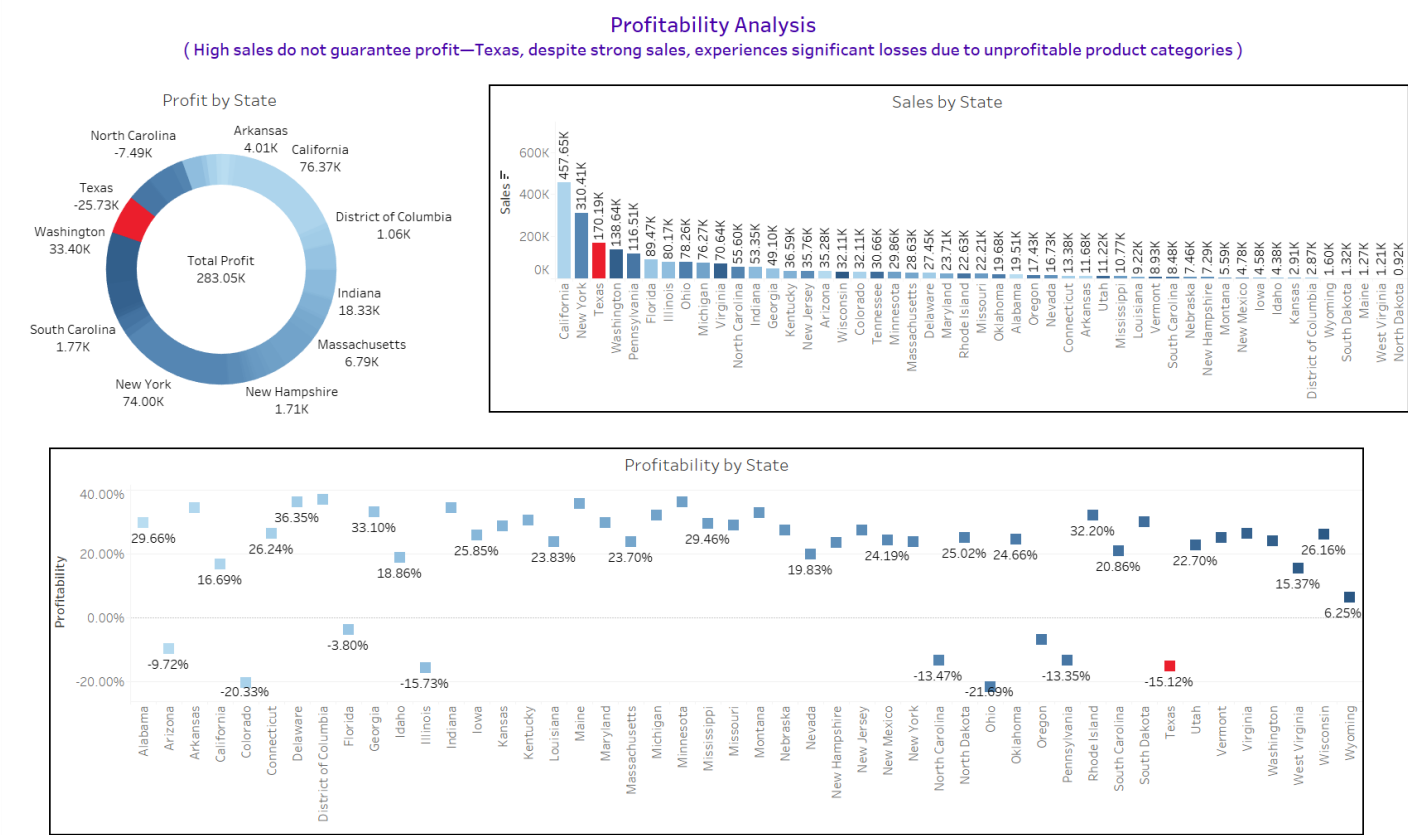
The scatter plot indicates the correlation of profit (x-axis) and sales (y-axis) across various sub-categories. This enables users to identify high and underperforming sub-categories rapidly. For instance, copiers and phones generate high sales and profitability whereas tables exhibit low profit margins due to pricing or cost issues.

1.4 Profit Analysis by Segment and Category



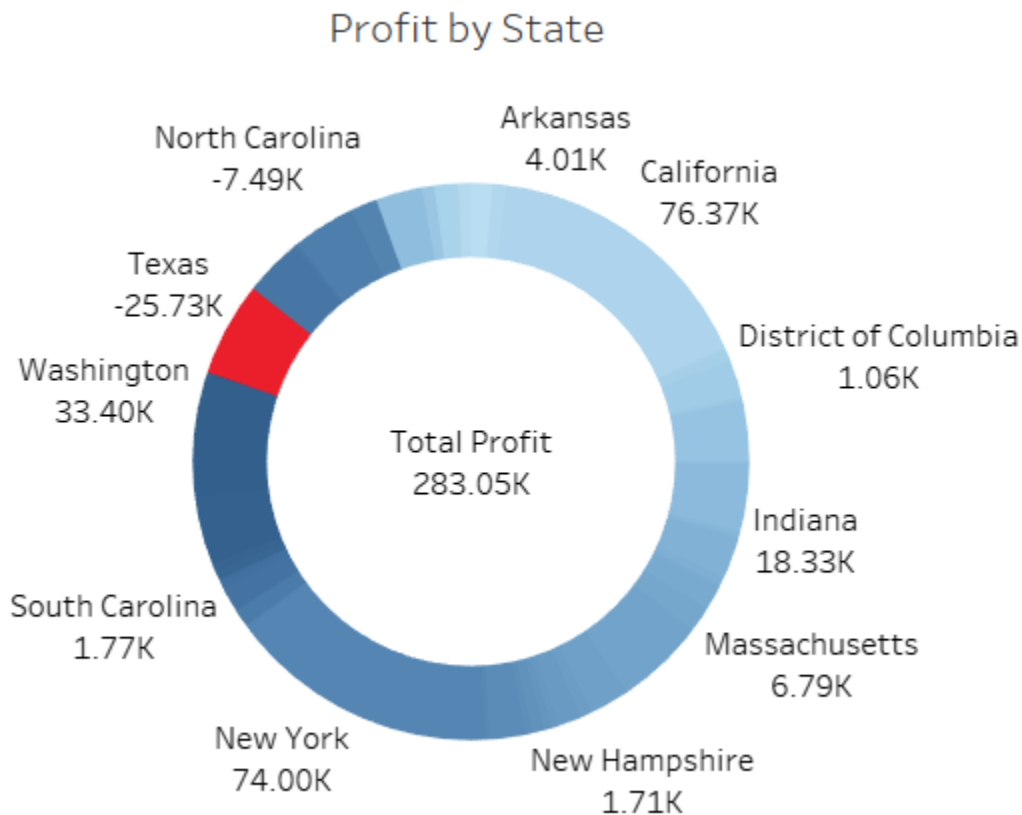
This horizontal bar chart displays profit distribution across different segments and product categories. Bar length and color indicate profit allocation, highlighting the most profitable area. For example, the consumer segment leads in all especially in the technology category. However, certain underperforming categories may contribute to state-level losses, as seen in Figure 2.

Figure 2:



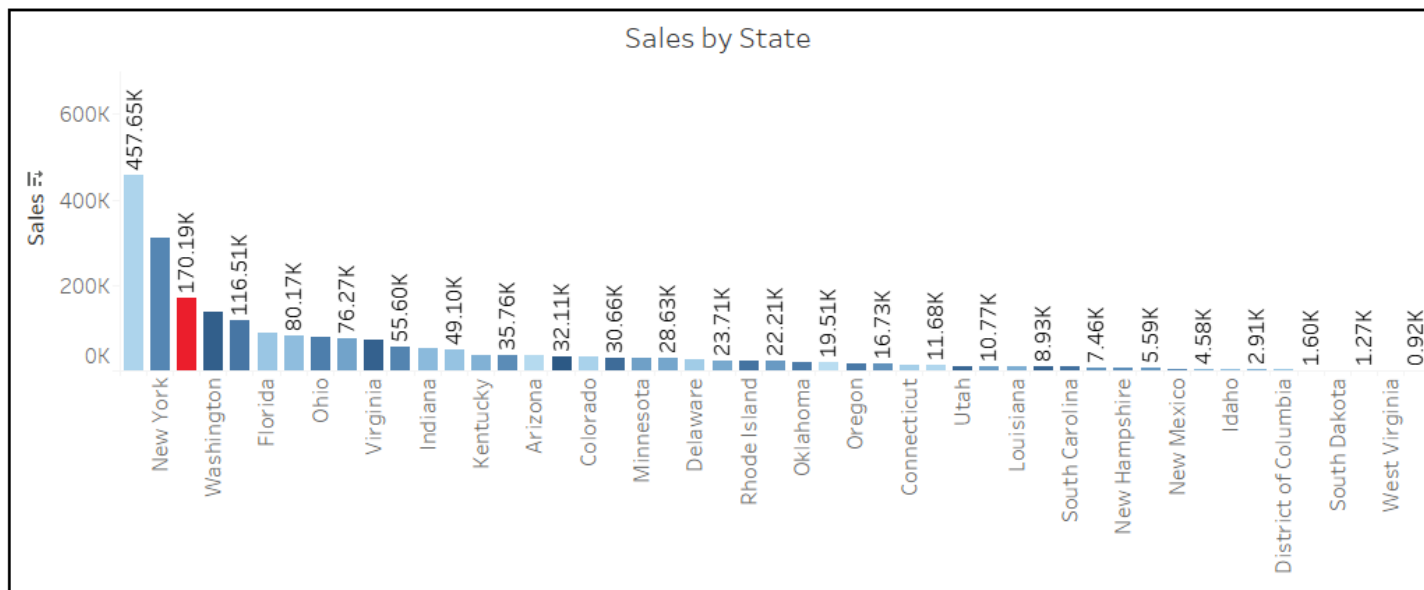
The previous figure shows that the U.S. is the major leader in profit and sales. However, significant losses in certain states impact the country’s overall profit sustainability. This dashboard helps to visualize these issues by focusing on the state-level performance.

2.1 Profit contribution by State



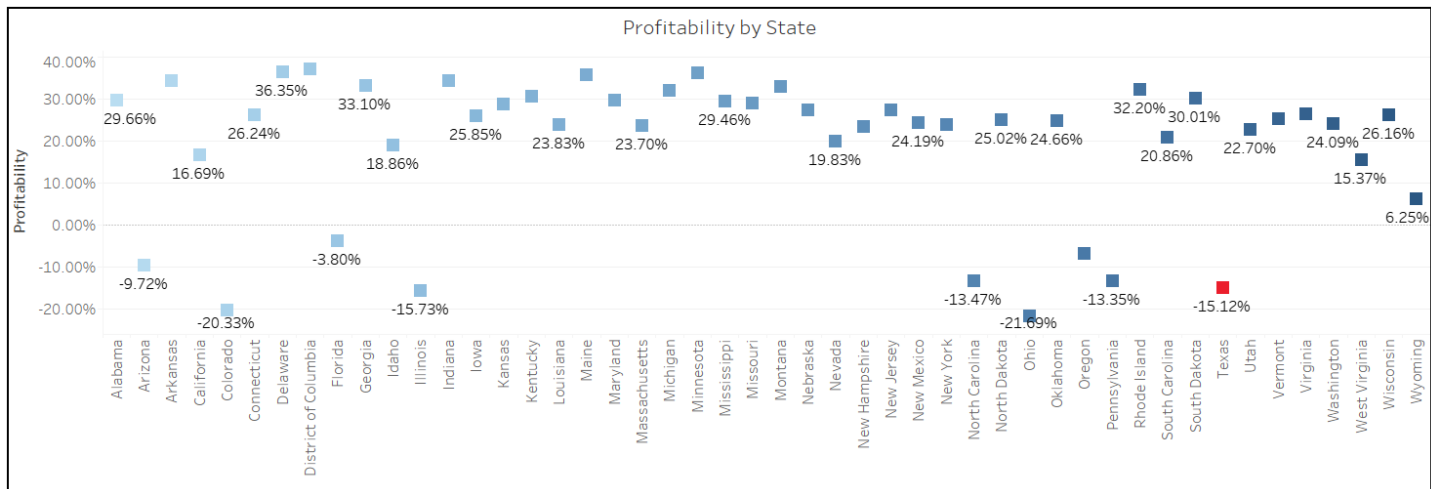
The donut chart indicates the distribution of total U.S. profit among states. California and New York have respectively contributed \$76.37K and \$74K to the overall profit. In contrast, Texas negatively impacts the overall sustainable profitability with tremendous losses of \$-25.73K in the United States. Thus, it is important to discover the root causes of this negative trend in Texas.

2.2 State Sales Volume



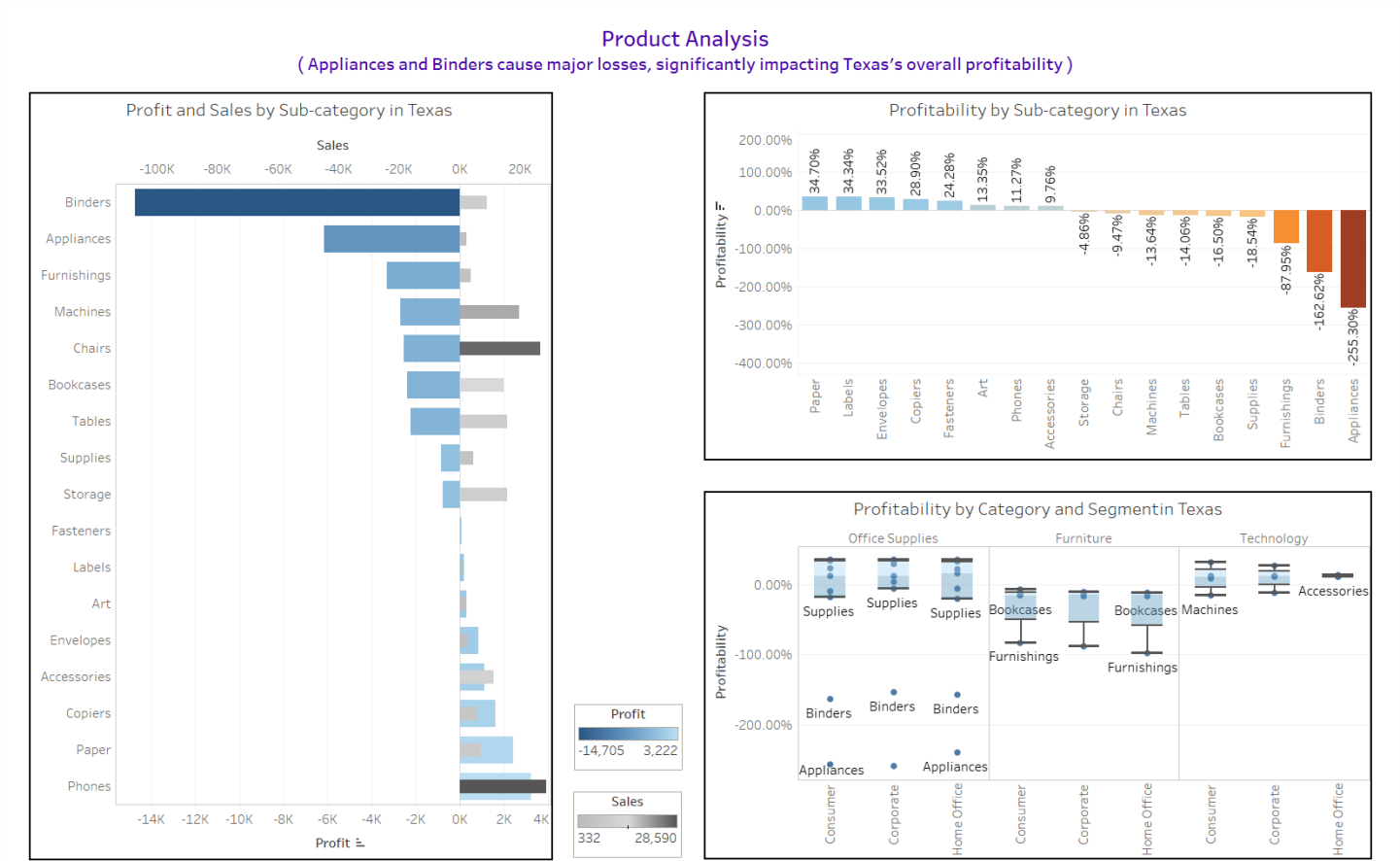
This bar chart ranks states by their sales volume to make it easy to compare how each state is performing. The donut chart and this bar chart show California and New York generate high sales and profits. However, Texas, which is in the top 3 for sales, has negative revenue. This proves that there are some serious issues in Texas that need attention.

2.3 State Profitability Analysis



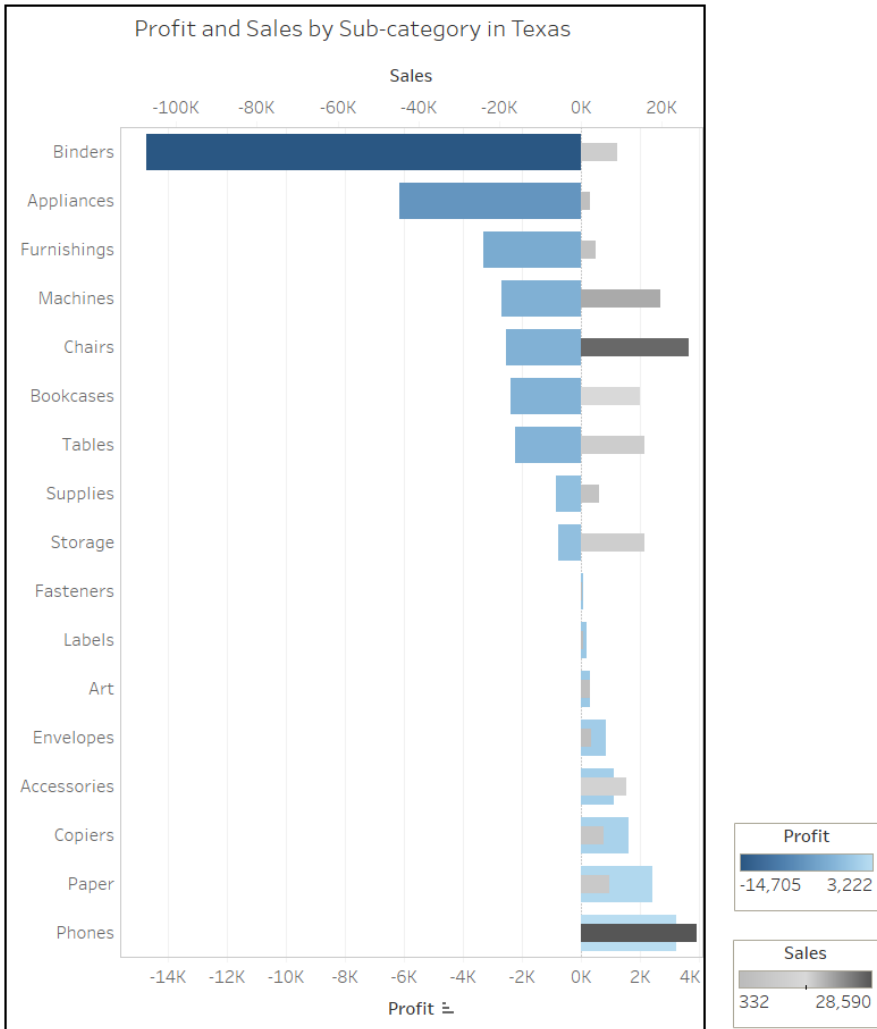
This scatter plot presents the U.S. profitability with profitability on the x-axis and states on the y-axis. It outlines financial disparities: profitable states like Delaware (36.35%) versus losses in Texas (-15.21%). Texas, despite its high sales ranking, experiences losses due to unprofitable product categories. This emphasizes that strong sales do not ensure profitability. The following figure shows the product impacting on Texas’s overall profitability.

Figure 3:



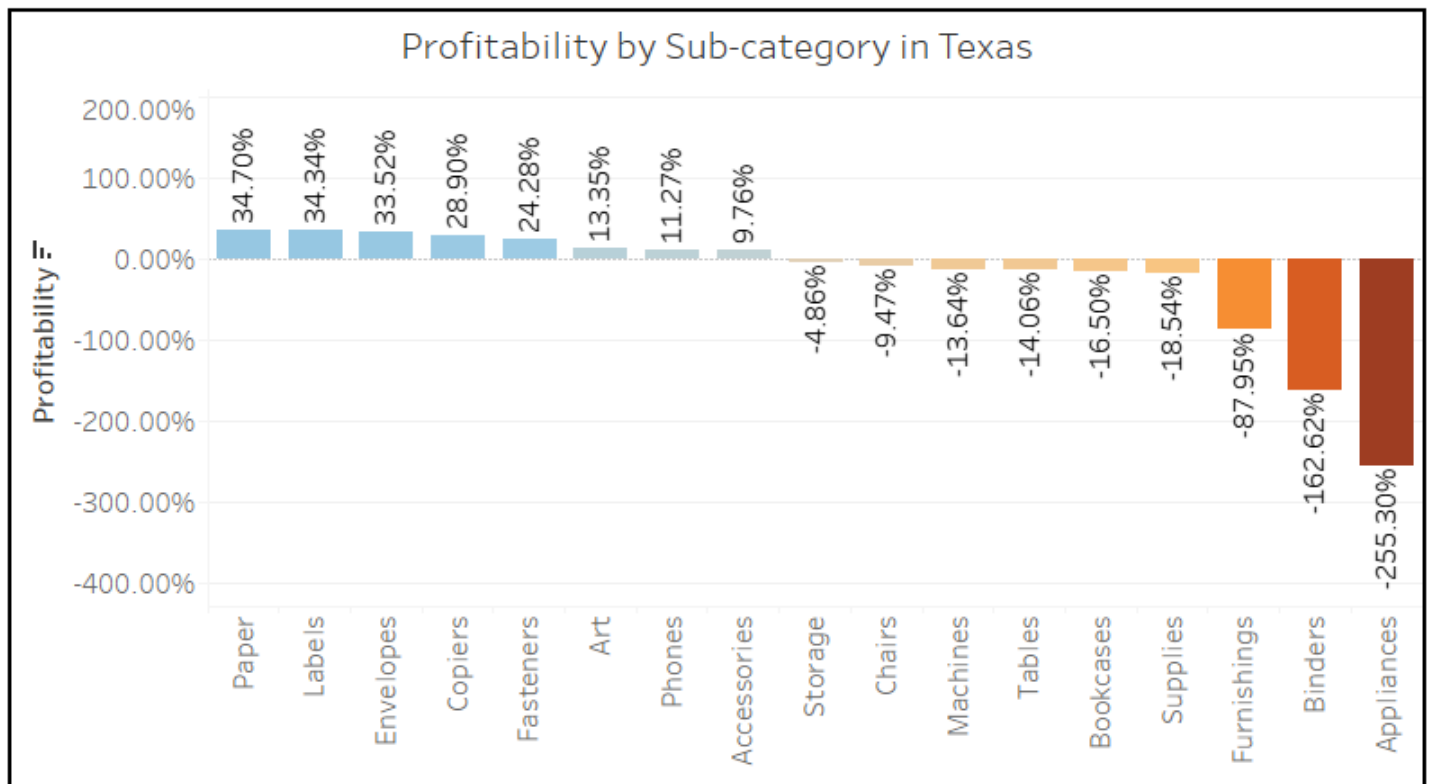
By analyzing the profitability across sub-categories, this figure identifies binders and appliances are the main causes of financial losses that substantially lower Texas's overall profit.

3.1 Profit and Sales by Sub-category



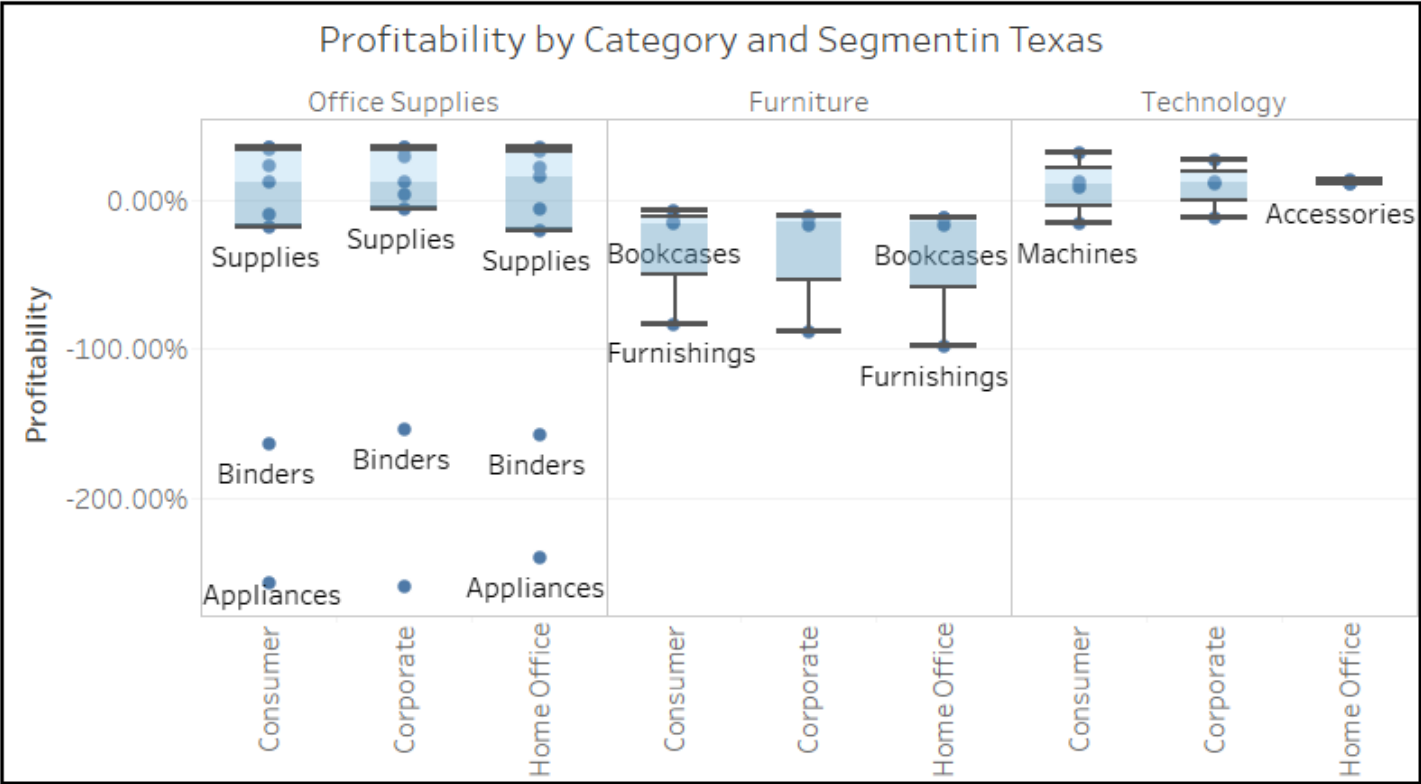
This dual-axis bar chart displays the profit and sales performance of sub-categories in Texas. Additionally, it indicates which products’ sales and profits are out of proportion. Binders and Appliances show substantial losses because of their negative revenue while products like phones show positive profitability which boosts the total profit. This chart is important for exploring which sub-categories that drag down profit.

3.2 Profitability Across Sub-categories



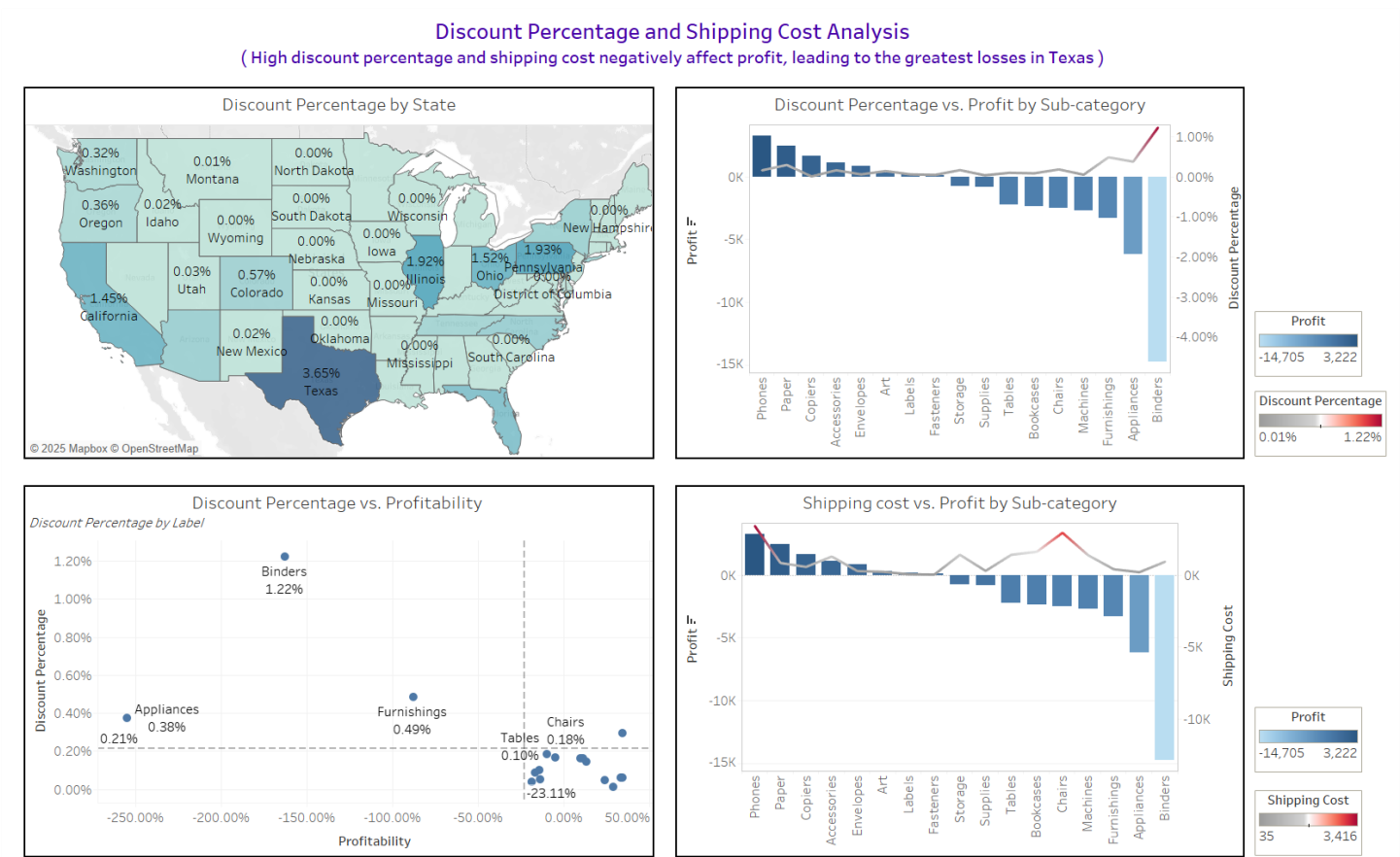
The goal of this bar chart is to identify the most unprofitable product by visualizing the profitability percentage of several sub-categories. While products like appliances (-255.30%) and binders (-162.62%) stand out as the most unproductive items reducing Texas's overall profit, paper (34.70%) and labels (34.34%) perform well in profitability percentage. Therefore, the primary issues causing Texas's poor profitability are binders and appliances.

3.3 Profitability by Product Category and Customer Segment



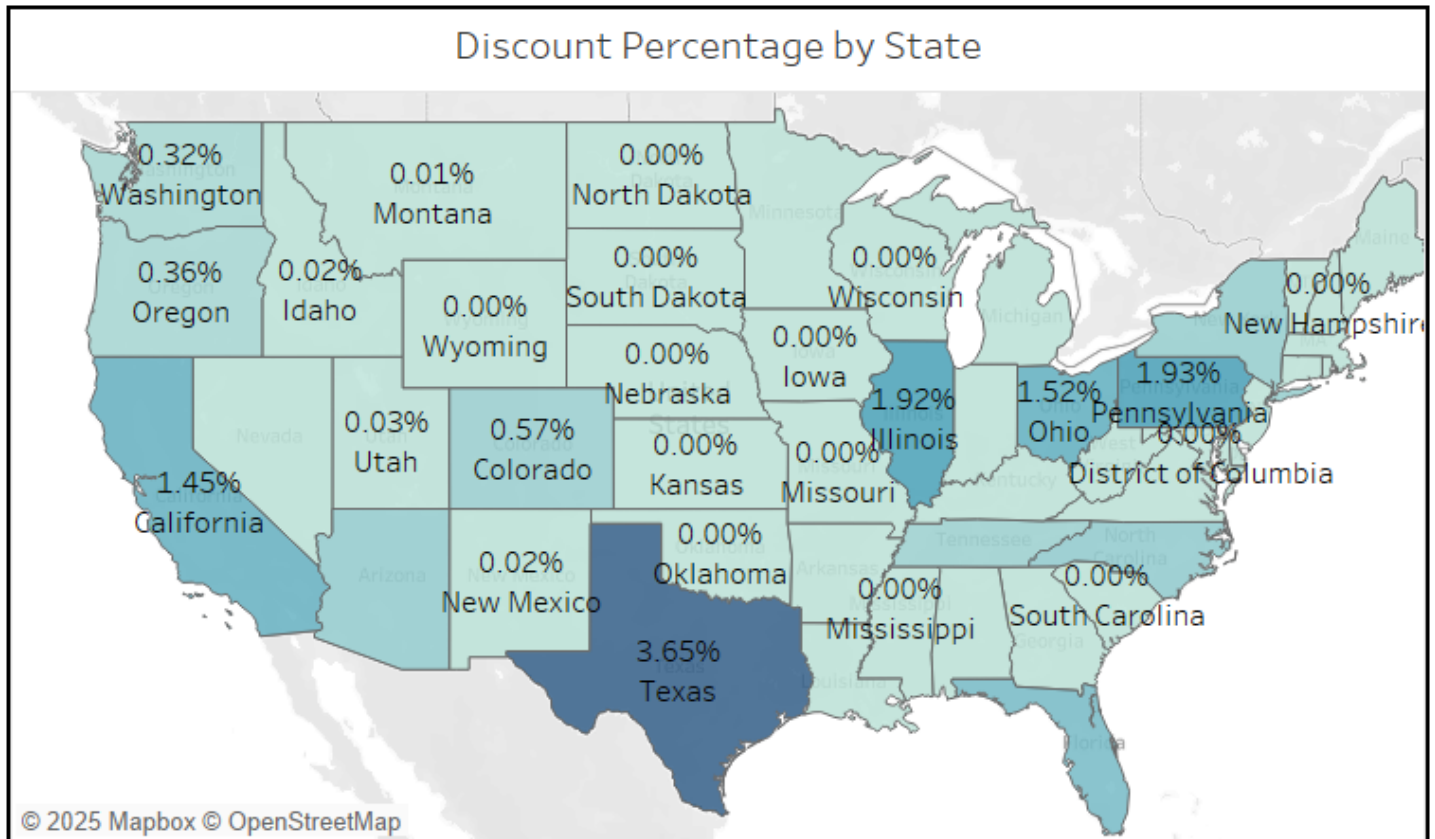
This box plot shows the profitability performance across each segment and category. Binders and appliances fall under the category of office suppliers and show consistent losses as outliers across all segments. These emphasize the negative trends of unprofitable items that require attention. The primary reasons behind these losses are the high discount percentage and shipping costs which will be further analyzed in the next figure.

Figure 4:



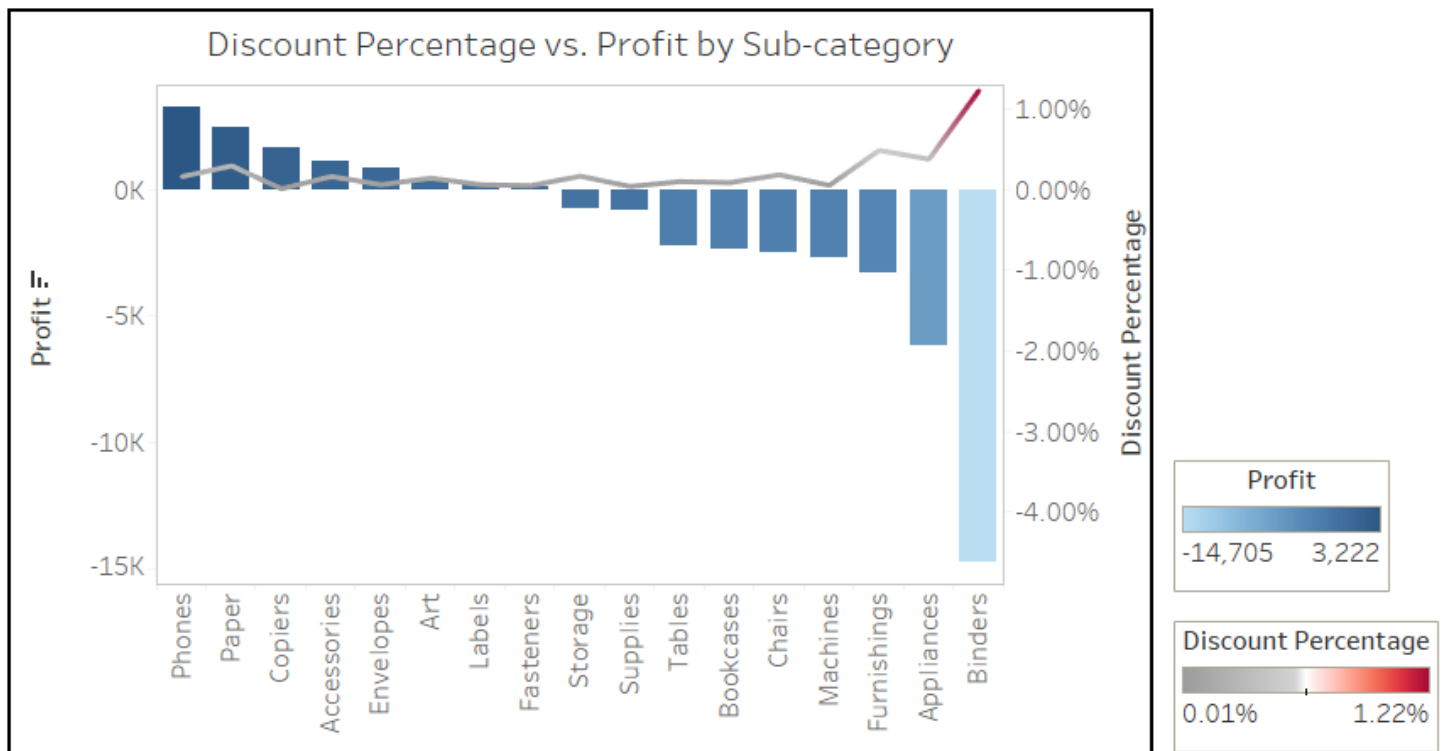
This figure analyzes how excessive discount percentages and inflated shipping costs affect profitability which leads to the detrimental effect in Texas and demonstrates the correlation between these factors and profit declines.

4.1 Geographic Discount Percentage Distribution



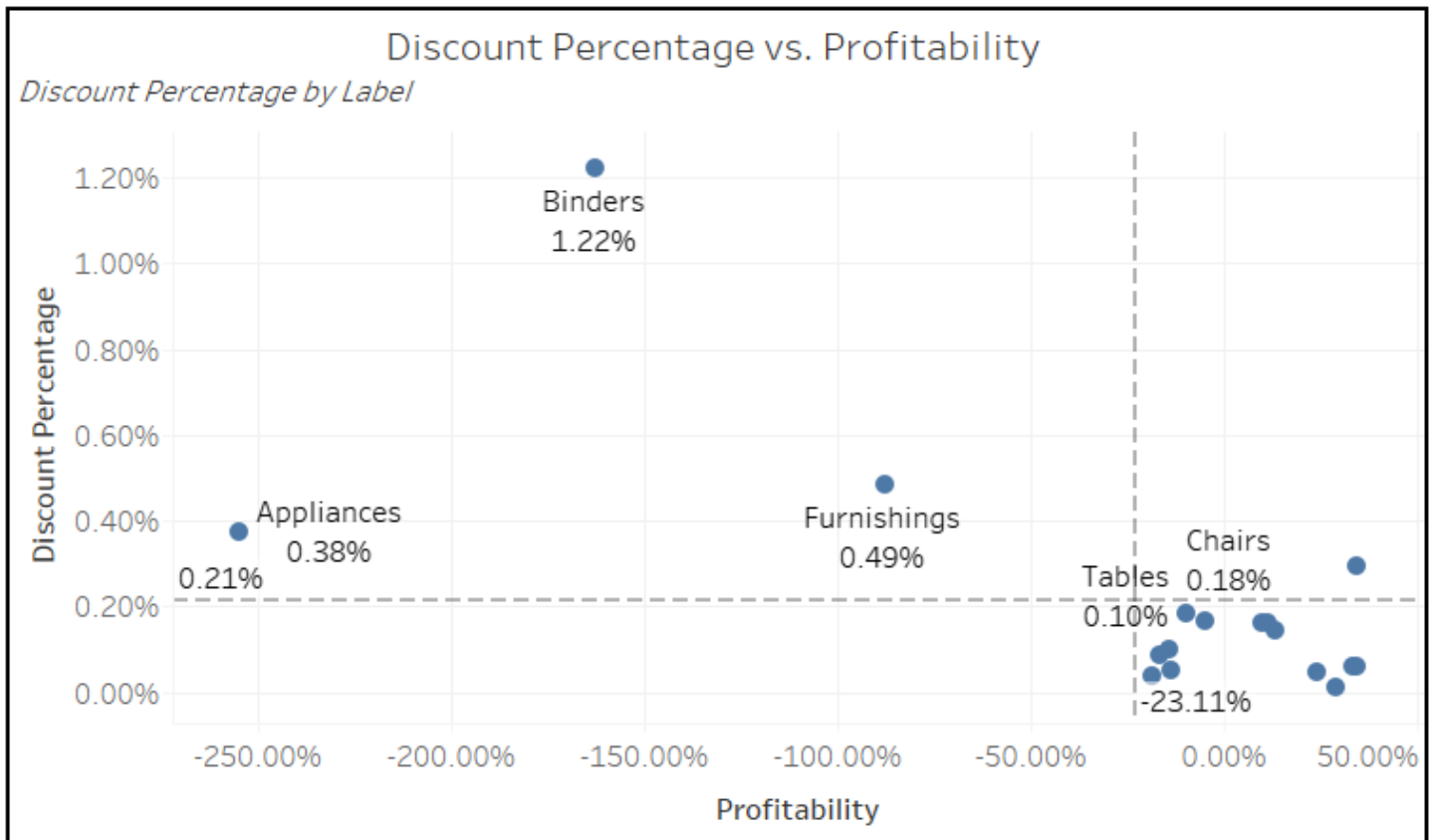
This map graph illustrates the discount percentages implemented by each state. Texas has the darkest area that indicates the greatest discount percentage (3.65%) among other states, strongly linked to profit decreases. Other states like Pennsylvania (1.93%) and Illinois (1.92%) also have relatively high discounts thus they tend to have poorer profitability which can be referred to in chart 2.3.

4.2 Correlation of Discount and Profit



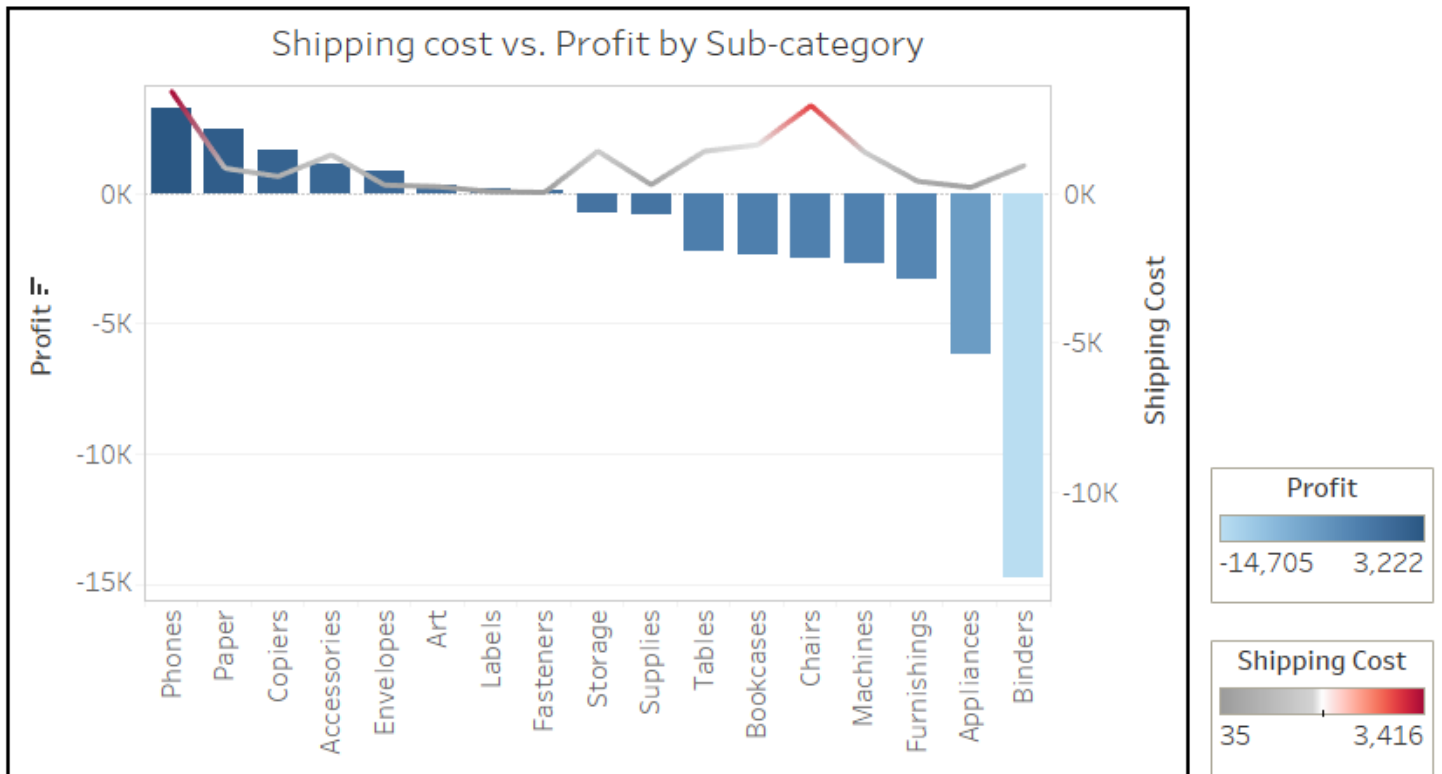
This chart compares discount percentage (line) and profit (bars) across different sub-categories. Several products show a negative correlation which indicates that higher discount rates lead to lower profitability, particularly binders and appliances. Binders have the highest discount rate but also the lowest profit, demonstrating a disproportionate relationship. Hence, a pricing strategy may be necessary.

4.3 Relationship between Discount Percentage and Profitability



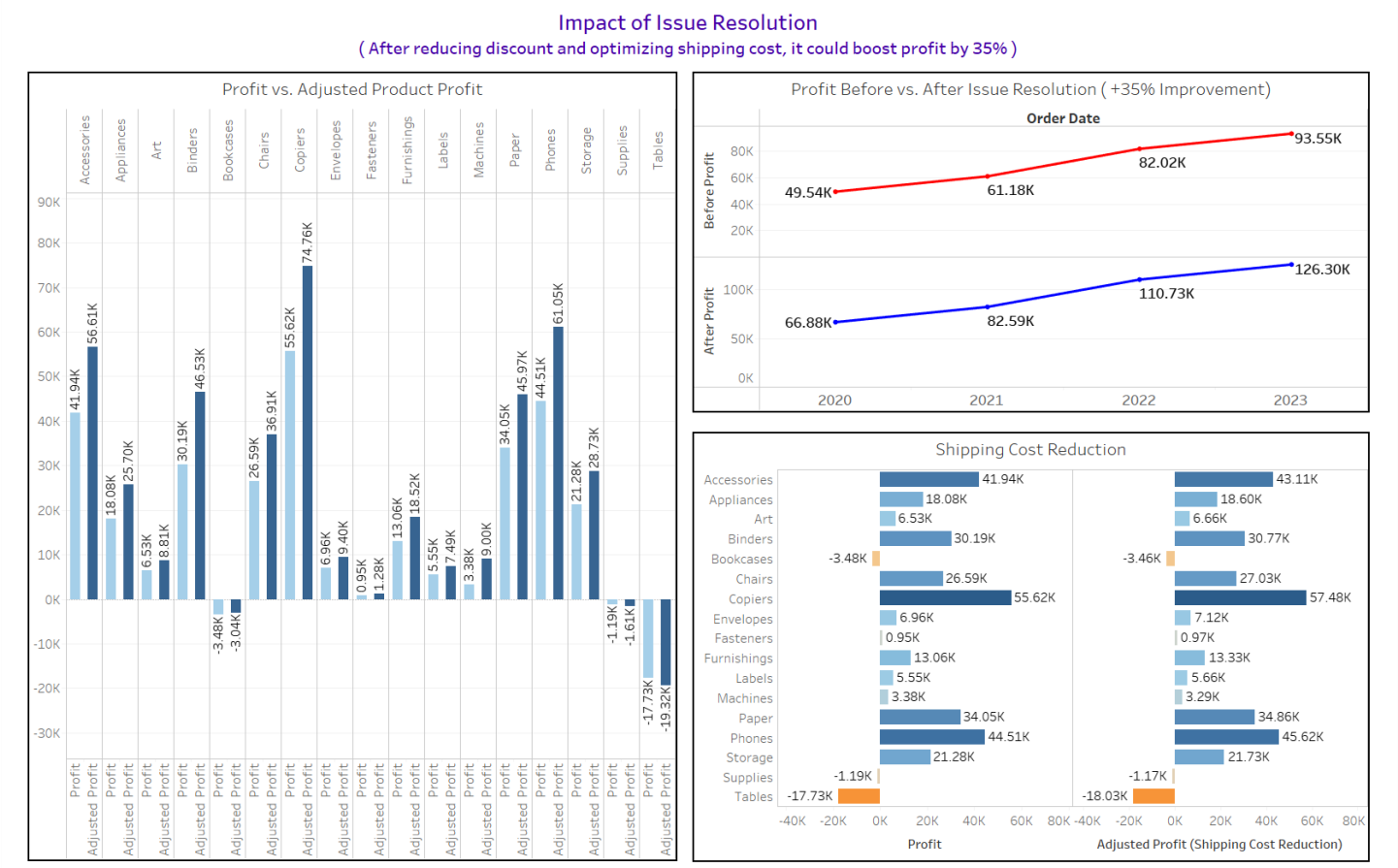
The scatter plot shows that binders stand out with a high discount percentage (1.22%) and low profitability. Appliances (0.38%) and Furnishings (0.49%) also show losses with above-average discount percentages. Conversely, chairs and tables have lower discount percentages and positive profitability. Due to many sub-categories clustering near 0% profitability, Texas faces difficult challenges in achieving profitability.

4.4 Correlation of Shipping Cost and Profit



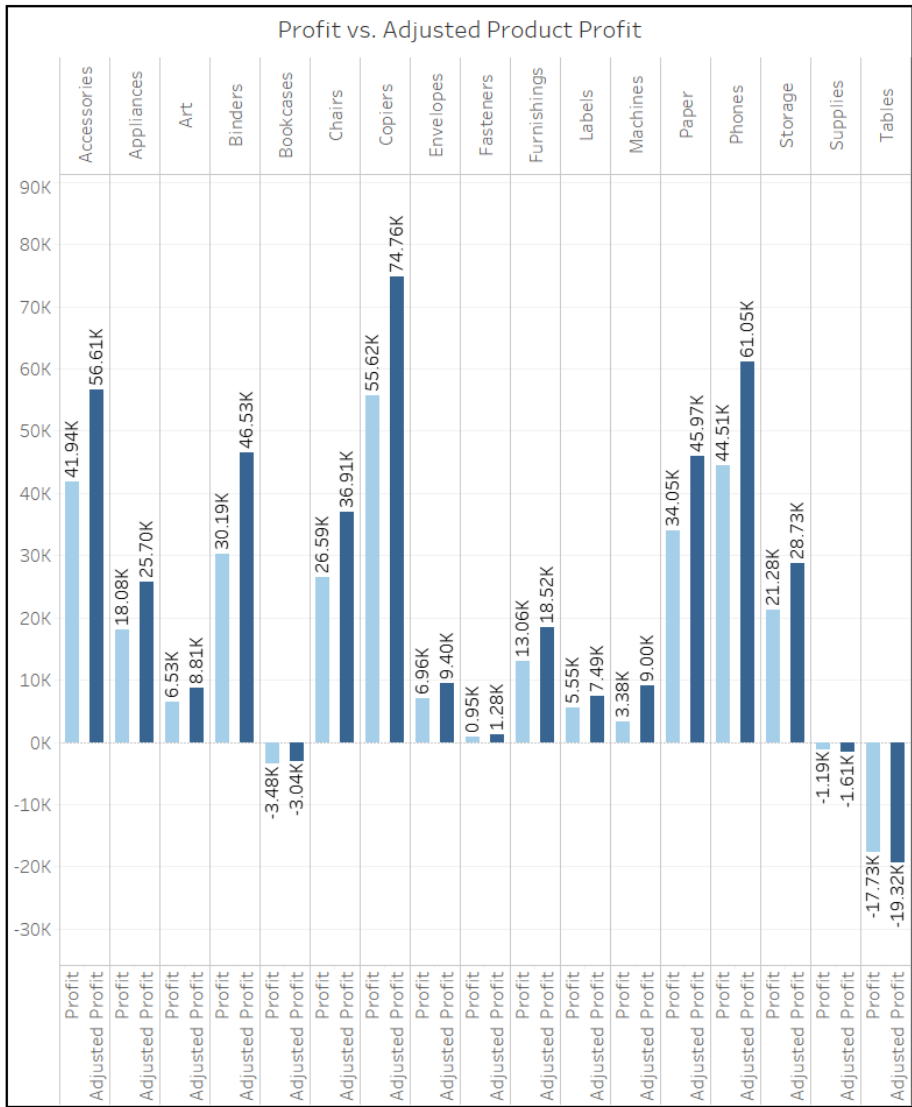
Inflated shipping costs tremendously impact profitability, especially bulky items like chairs and vulnerable items like phones. Therefore, it is essential to optimize cost. Reducing discount rates and optimizing shipping costs could significantly lead to substantial improvement in overall profit as reflected in the final figure.

Figure 5:



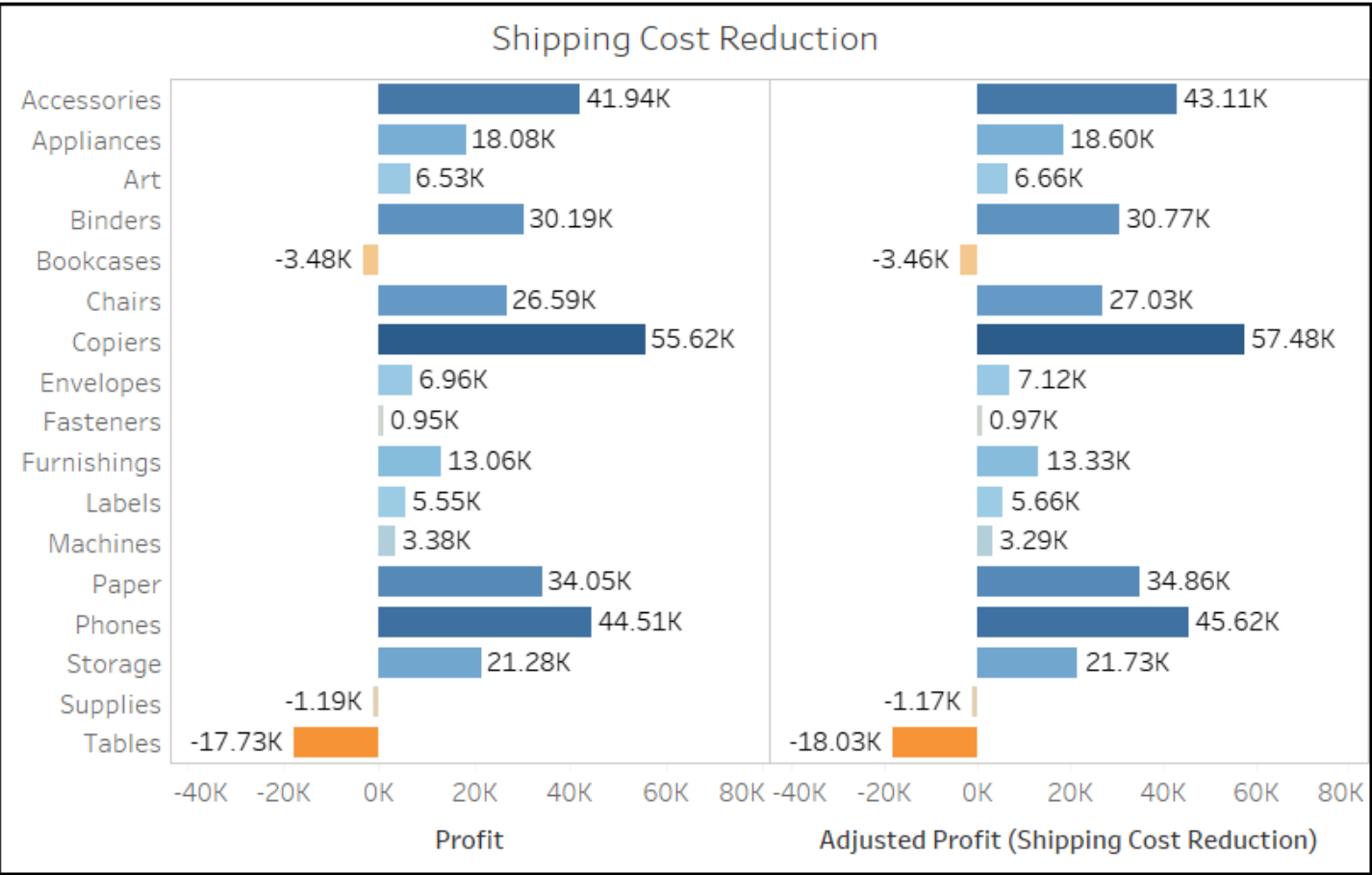
This dashboard visualizes the implication of adjustments and presents a 35% profit improvement achieved by optimizing the unprofitable factors.

5.1 Profit Comparison before and after Discount Optimization



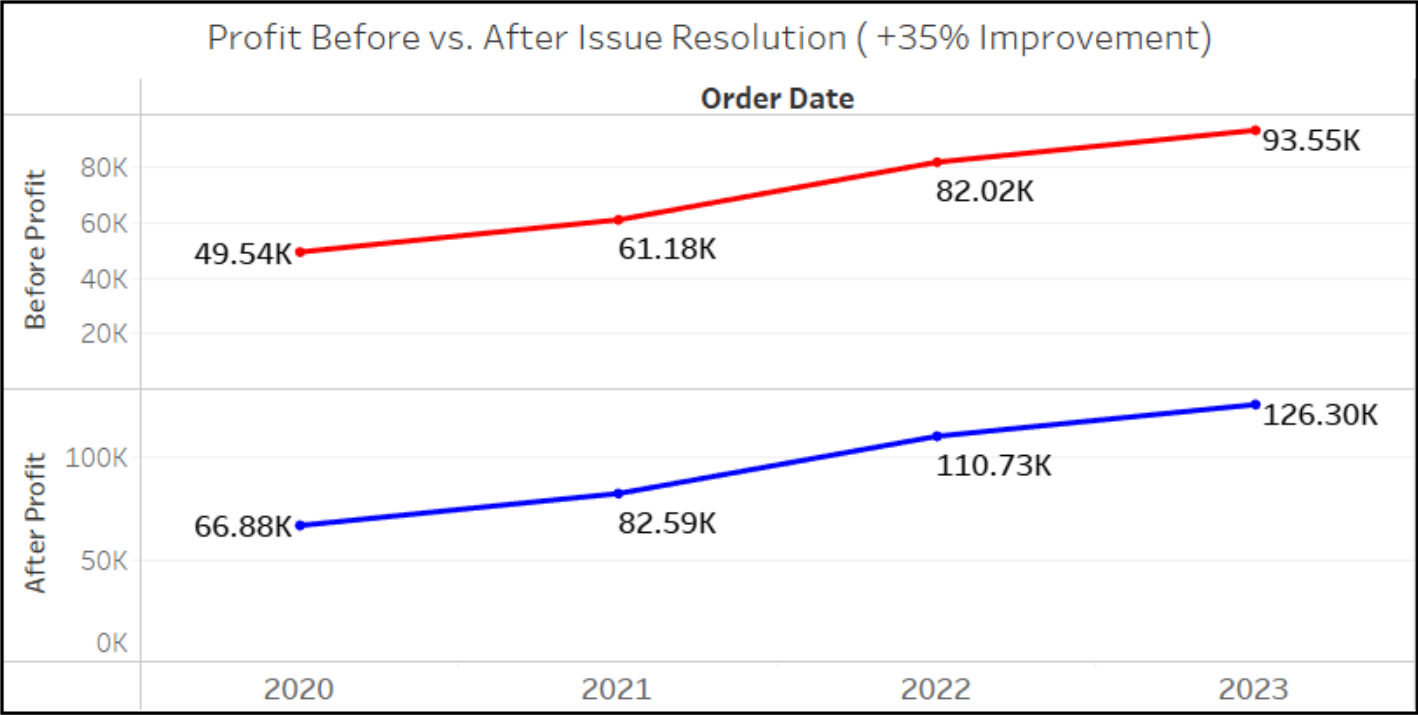
This side-by-side bar adjusts profit based on discount percentage level. For products with discount percentages above 20%, profit increases by 20%. This is because reducing high discount percentages raises the selling price while costs remain unchanged which improves profit margins. For discount percentages below 20%, profit is increased by 35% since the products are already profitable and become more profitable with lower discount percentages. However, some products like tables and bookcases still underperform but they reduce the profit loss.

5.2 Profit Comparison before and after Shipping Cost Reduction



This chart illustrates how negotiating shipping costs based on shipping mode helps businesses retain more profit. Premium shipping options like same-day and first-class shipping incur higher costs that decrease profitability. Same-day shipping is the most expensive so optimizing it can increase profit by 15%. First-class shipping is slightly cheaper, potentially boosting profit by 10%. Meanwhile, standard and second-class shipping remain unchanged as their cost-effectiveness. This method ensures that revenue covers shipping expenses to prevent losses from premium delivery services.

5.3 Profit Trend After Issue Resolution



This line chart depicts the yearly profit trend in the U.S. before and after adjusting discount percentages and shipping costs. The purpose of this analysis is to evaluate the effectiveness of these adjustments. The average difference between original profit (red) and adjusted profit (blue) is approximately \$25.05K. While the original profit grows by about \$14.67K annually, the adjusted profit increases by about \$19.81K per year. By 2023, the adjusted profit reaches \$126.30K compared to \$93.55K before adjustments showing a potential 35% profit growth.

Recommendation

A combination of discount reduction and shipping cost optimization is more effective than using only one strategy to maximize profitability and maintain competitive pricing at the same time. In Texas, the product portfolio should be reconsidered. Underperforming items like binders and appliances can be phased out. Furthermore, concentrating on high margin products like phones and paper can further increase profitability. Moreover, targeted markets and customer groups should be developed to ensure suitable products are brought to the right person. There are more strategy can be explored like pricing strategy which can help to maintain customer satisfaction, leading to sustainable profit improvement.

Conclusion

Dashboard 1 shows a widening gap between revenue and sales volume in the U.S., indicating severe issues that need to be resolved. Dashboard 2 analyses state profitability and identifies Texas as a main concern with high sales but low profit, proving that high sales do not ensure high profitability. The following dashboard further explores Texas to disclose binders and appliances are the primary unprofitable sub-categories that cause poor profitability. Dashboard 4 indicates high discount percentages and excessive shipping costs are the root cause of low profitability in Texas. Finally, dashboard 5 presents the outcome after these factors. In conclusion, the business objective of a 35% profit improvement in the United States has been successfully achieved.